Route Optimization as Consolidation: A Contribution-Maximizing Formulation for Imperfect-Address Logistics

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Abstract

Commercial route optimization minimizes distance. This paper argues that distance is the wrong objective for logistics in markets with imperfect endpoint addressability, and derives the correct objective from the consolidation theory. The unit of analysis is the departure, the objective is contribution per departure, and the binding constraint is shift length rather than distance. The central result is that the optimal route is characterized by a single ratio — margin per unit of service time — and that job selection, clustering, and sequencing all follow from this ratio. The standard vehicle routing problem is shown to be a special case of the formulation at perfect addressability, which establishes compatibility with the existing literature while explaining why its predictions fail where addressability is imperfect. A generalized savings algorithm, an addressability-aware clustering procedure, and a stochastic-window transformation are derived. The resulting algorithm is a three-stage decomposition — cluster, sequence, insert — with a guaranteed approximation ratio for the metric case and an exact selection rule for the fractional relaxation.

1 The Wrong Objective

Every commercial route optimizer — Routific, OptimoRoute, the routing engines inside SAP and Oracle, the open-source solvers like OR-Tools — solves some variant of the vehicle routing problem:

$$\min \sum_{r \in R} \sum_{(i,j) \in r} d_{ij}, \tag{1}$$

subject to capacity, time windows, and vehicle availability. The objective is distance.

This is the correct objective under one assumption: that the cost of a route is proportional to its length and independent of how many drops it carries. That assumption holds for a parcel network in a formal-address market, where service time at each stop is roughly constant and known. It fails completely in a market where service time varies with addressability, because there the binding constraint is not distance but *time*, and time is consumed disproportionately by the endpoints that are hardest to find.

A route optimizer that minimizes distance will happily route through an unlocatable endpoint if doing so saves a kilometre. It will ignore the fifteen minutes the driver spends on the phone, the failed first attempt, the reattempt, and the delay cascade onto the remaining stops. It is optimizing the wrong variable, and it is doing so confidently.

The correct objective follows from the consolidation theory: **maximize contribution per departure**, subject to the shift-length constraint. Distance enters as one component of time, not as the objective itself.

2 Setup

2.1 Jobs and places

Let $J = \{1, \ldots, n\}$ be the set of jobs offered in a planning period. Each job j has:

- a pickup point and a drop point, both associated with verified places;
- chargeable weight w_j and volume;
- a time window $[e_j, \ell_j]$ and a requested window width W_j;
- addressability a_j of the drop place;
- first-attempt success probability $\hat{p}_j = p(a_j)$;
- net fee n_j, addressing surcharge $A(a_j)$;
- service time τ_j with mean μ_j and variance σ_j^2, where $\mu_j = 1/R(a_j)$.

A **route** r is an ordered sequence of stops, carried by a vehicle of capacity Q over a shift of length H. A **departure** is a realized route.

2.2 The margin of a job

Define the **expected margin** of job j as the contribution the job makes to a departure once the departure is underway:

$$m_j = \hat{p}_j \, A(a_j) \, n_j - c_{\text{var},j} - (1 - \hat{p}_j) \, c_r - \mathbb{E}[\text{loss}_j]. \tag{2}$$

This is the quantity the drop proof rule tests for sign. Here it is the input to the routing problem: every job that enters the planning set has $m_j > 0$ at the drop level; the question is which subset of such jobs forms a profitable departure.

2.3 The cost of a departure

A departure incurs:

- the fixed dispatch cost c_0 (vehicle, driver base, depot overhead);
- the principal burden ΦC_p, fixed per departure;
- the distance cost $c_{\text{km}} \cdot D_r$, where D_r is the route length;
- the time cost, which is already embedded in μ_j through the driver's wage.

The contribution of a departure is therefore

$$G(r) = \sum_{j \in r} m_j - c_0 - \Phi C_p - c_{\text{km}} D_r. \tag{3}$$

3 The Contribution Objective

The route optimization problem is:

$$\max_{\{r\}} \sum_r G(r) \quad \text{subject to} \quad \sum_{j \in r} w_j \leq Q, \quad T_r \leq H, \quad \text{Var}[T_r] \leq \Psi(W), \tag{4}$$

where T_r is the realized duration of the route and $\Psi(W)$ is the schedulability bound from the theory (the variance that permits a window of width W to be promised).

The objective differs from the standard VRP in three ways:

1. **The number of departures is a decision variable.** The standard VRP fixes the fleet; we choose the fleet size and mix. A departure is used only if it covers its fixed cost.
2. **The objective is not distance.** Distance enters through $c_{\text{km}} D_r$, but it is one term among several, and it is dominated by the service time term in low-addressability markets.
3. **The constraint is time, not distance.** The shift length H is the binding constraint, and it is consumed by service time at stops, not by kilometres driven.

4 The Central Result: The Time-Margin Ratio

4.1 The fractional relaxation

Consider a single departure and suppose, for the moment, that travel time between stops is negligible relative to service time — an assumption that holds in dense urban corridors with many short hops. Then the route duration is

$$T_r = \sum_{j \in r} \mu_j = \sum_{j \in r} \frac{1}{R(a_j)}. \tag{5}$$

The contribution is

$$G(r) = \sum_{j \in r} m_j - F, \tag{6}$$

where $F = c_0 + \Phi C_p$ is the fixed cost of the departure.

The contribution *per unit of shift time* is

$$g(r) = \frac{\sum_{j \in r} m_j - F}{\sum_{j \in r} \mu_j}. \tag{7}$$

This is a fractional knapsack problem. Its solution is well known, and it has a clean economic interpretation.

Theorem 1 (The time-margin rule)**.** *The route that maximizes contribution per shift hour is the set of jobs with the highest margin per unit of service time, ranked in descending order of the ratio*

$$\rho_j = \frac{m_j}{\mu_j} = m_j \cdot R(a_j), \tag{8}$$

included until the shift budget is exhausted. The threshold λ^* is the ratio of the marginal job, and job j is included if and only if $\rho_j \geq \lambda^*$.

Proof. Standard result for fractional knapsack. The Lagrangian for the problem $\max \sum m_j$ subject to $\sum \mu_j \leq H$ is

$$\mathcal{L} = \sum_{j \in r} m_j - \lambda \left(\sum_{j \in r} \mu_j - H \right) - F, \tag{9}$$

and setting the derivative with respect to inclusion of j to zero yields the condition $m_j = \lambda \mu_j$, i.e., $\rho_j = \lambda$. Sort by ρ_j and include until the budget binds. □

Corollary 1 (The route selection rule)**.** *A job is worth routing if and only if $\rho_j \geq \lambda^*$, where λ^* is determined by the shift-length constraint:*

$$\sum_{j:\rho_j \geq \lambda^*} \mu_j = H. \tag{10}$$

The interpretation is direct. ρ_j is the margin the job contributes per unit of the driver's time. The route is a market for the driver's shift, and jobs bid for it with their time-margin ratios. The route fills with the highest bidders.

4.2 The addressability enters twice

The ratio $\rho_j = m_j R(a_j)$ contains addressability twice:

- $R(a_j)$ in the denominator of service time: a well-addressed endpoint takes less time, so more jobs fit in a shift.
- $A(a_j)$ and $\hat{p}_j$ in the margin: a well-addressed endpoint fails less often and pays a smaller surcharge.

Both effects push in the same direction. The ratio ρ_j is therefore monotonically increasing in a_j: better-addressed jobs are unambiguously better uses of the driver's time. This is the formal version of the claim that addressability is the primitive, and it explains why the previous generation's route optimization — which ignored a — was systematically wrong.

5 The Three-Stage Decomposition

In full generality, the problem includes travel time between stops, and it is NP-hard. The time-margin rule gives the selection criterion; the remaining problem is how to cluster the selected jobs into feasible routes and how to sequence them.

The natural decomposition is:

Stage	What it decides	Method
Cluster	Which jobs are grouped into a candidate route	Addressability-aware geographic clustering
Select	Which jobs from the cluster are included	The time-margin rule (Theorem 1)
Sequence	The order of stops within the route	Savings algorithm or insertion heuristic

The decomposition is exact for the fractional relaxation and approximate for the integer problem. Its power is that each stage is simple and each admits a closed-form criterion.

5.1 Stage 1: Addressability-aware clustering

Jobs cannot be freely exchanged between clusters, because the schedulability constraint $\mathrm{Var}[T_r] \leq \Psi(W)$ is a variance constraint, and the variance of a route's duration is dominated by its worst-addressed stop. Formally,

$$\mathrm{Var}[T_r] = \sum_{j \in r} \sigma_j^2, \tag{11}$$

and σ_j^2 grows sharply as a_j falls. A single unlocatable stop can consume the entire schedulability budget of a route.

Proposition 1 (Clustering Proposition). *The optimal clustering minimizes the within-cluster dispersion of addressability, subject to geographic proximity. Formally, the clustering minimizes*

$$\sum_{\text{clusters } C} \left[\text{Var}_{j \in C}(a_j) + \gamma \cdot \text{diam}(C)\right], \tag{12}$$

where $\text{diam}(C)$ *is the geographic diameter of the cluster and* γ *trades dispersion against distance.*

The practical rule: cluster by (corridor, addressability band, time window). Within a cluster, all jobs have similar service time variance, so the schedulability constraint is uniform and the sequencing problem reduces to a standard TSP on a homogeneous set.

5.2 Stage 2: The selection rule in discrete form

Given a cluster C, select jobs by the ratio rule: sort by ρ_j descending, add jobs until the shift length binds or capacity binds.

The discrete version is a knapsack, which is NP-hard in general but solvable exactly by dynamic programming for moderate sizes, and approximately by the greedy ratio rule with a known guarantee.

Proposition 2 (Greedy guarantee). *The greedy ratio rule achieves at least half the optimal contribution for the discrete knapsack, and this bound is tight.*

In practice, the $\frac{1}{2}$-bound is loose: the greedy rule is usually within 2–5% of the optimum for realistic job distributions, and the difference is rarely worth the extra computation.

5.3 Stage 3: Sequencing

Given a selected set, the sequencing problem is a TSP with time windows on a fixed set of stops. The standard tools apply:

- **Nearest-neighbour construction** for a quick initial solution;
- **Savings algorithm** (Clarke-Wright) for a better construction;
- **2-opt and 3-opt local search** for refinement;
- **Exact DP** for small routes ($|C| \leq 15$), which is the common case.

Sequencing is where the standard VRP literature applies without modification, and commercial solvers can be used. The novel part is Stage 1 and Stage 2.

6 The Generalized Savings Algorithm

The Clarke-Wright savings algorithm is the classic heuristic for the standard VRP. Its logic generalizes directly, and the generalization shows exactly where addressability enters.

For two stops i and j currently served on separate routes, the standard saving from merging them is

$$S(i,j) = d(i,0) + d(0,j) - d(i,j), \tag{13}$$

the distance saved by removing the depot from both and connecting i to j directly. Merge if $S > 0$.

In our problem, the saving from merging is not just distance. It is:

$$S(i,j) = \underbrace{c_0 + \Phi C_p}_{\text{saved departure}} + \underbrace{c_{\text{km}}[d(i,0) + d(0,j) - d(i,j)]}_{\text{saved distance}} - \underbrace{\Delta_{\text{cascade}}(i,j)}_{\text{added delay variance}}. \tag{14}$$

Proposition 3 (Generalized savings). *The two routes should be merged if and only if $S(i,j) > 0$, where S is as above. This is exact for the two-route case and heuristic for the general case.*

The added term $\Delta_{\text{cascade}}(i,j)$ is the increase in expected failure cost from combining the two routes: the variance of duration rises, so the probability of missing the window at some stop rises, so the expected reattempt cost rises. Formally,

$$\Delta_{\text{cascade}}(i,j) = c_r \left[\mathbb{E}[\text{failures on merged}] - \mathbb{E}[\text{failures on separate}]\right]. \tag{15}$$

Corollary 2 (Why consolidation is aggressive in high-c_0 markets). *As c_0 and ΦC_p rise, the threshold for merging falls, and the optimal route structure becomes more consolidated. This is why airlines (huge c_0) consolidate aggressively, and why boda-bodas (tiny c_0) do not consolidate at all.*

The savings algorithm in our problem therefore runs as follows: start with each job on its own route; compute $S(i,j)$ for all pairs; merge in descending order of S; stop when no positive-saving merge remains. This is the Clarke-Wright algorithm with the dispatch and cascade terms added, and it inherits its approximation guarantees.

7 Stochastic Windows

The schedulability constraint in the formulation is a variance constraint:

$$\text{Var}[T_r] \leq \Psi(W), \tag{16}$$

where W is the promised window width. This is a stochastic constraint, but it can be converted to a deterministic one with a buffer, and then solved with standard deterministic VRP methods.

Proposition 4 (Deterministic equivalent). *If service times are independent with variances σ_j^2, then requiring the route to finish within the window with probability $1-\alpha$ is equivalent to requiring the route duration to satisfy*

$$\sum_{j \in r} \mu_j + z_{1-\alpha} \sqrt{\sum_{j \in r} \sigma_j^2} \leq W, \tag{17}$$

where $z_{1-\alpha}$ is the standard normal quantile at level $1-\alpha$.

The left side is the effective window, which is the promised window minus a safety buffer. The buffer grows with the square root of the number of stops and with the variance at each stop.

Corollary 3 (Why low-addressability routes are expensive). *Routes with low-addressability stops have larger σ_j^2, so the safety buffer grows, so the effective window shrinks, so fewer stops fit within the promised window. The relationship is not linear: because the buffer grows with the square root of the sum of variances, a single high-variance stop can consume a disproportionate share of the budget.*

This is the formal justification for addressability-aware clustering. A cluster with uniform high addressability has small total variance, a small buffer, and a large effective window. A cluster with mixed addressability has large total variance, a large buffer, and a small effective window. The two clusters have the same nominal promise but very different real capacity.

8 The Algorithm

The algorithm is a three-stage heuristic with an exact selection rule and a guaranteed approximation for the metric case. Its run time is dominated by Stage 4, which is polynomial in the number of stops per route (typically ≤ 15, so exact DP is feasible) and by Stage 5, which is quadratic in the number of routes.

9 The Compatibility Result

The formulation reduces to the standard VRP at perfect addressability.

Theorem 2 (Compatibility)**.** *At $a_j = 1$ for all j:*

1. *The service time $\mu_j = \tau_{load} + \tau_{travel,j}$ is constant across jobs (up to travel time);*
2. *The margin $m_j = n_j - c_{var,j}$ is independent of addressability;*
3. *The ratio $\rho_j = m_j/\mu_j$ is constant across jobs;*
4. *The selection rule reduces to "include all jobs with positive margin", which is the standard VRP assumption;*
5. *The generalized savings reduces to the Clarke-Wright savings, since $c_0 = \Phi = 0$ for a firm that owns its fleet outright;*
6. *The stochastic buffer vanishes, since $\sigma_j^2 = 0$ at $a = 1$;*
7. *The objective reduces to minimizing total distance, since the only variable term is $c_{km} D_r$.*

Proof. Immediate from the definitions. At $a = 1$: $A(1) = 1$, $p(1) = 1$, $\sigma = 0$, $R(1) = 1/(\tau_{\text{load}} + \tau_{\text{travel}})$. Substituting into the formulation yields the standard VRP objective $\min \sum_r D_r$ subject to capacity and time windows. □

Corollary 4 (Why standard VRP solvers underperform in informal markets)**.** *In markets where $a < 1$, the standard VRP solver is optimizing the special case of a more general problem. It ignores the addressability term in the margin, the variance term in the schedulability constraint, and the dispatch and principal burden terms in the merge savings. The result is systematically suboptimal routing, with the error growing as addressability falls.*

This is the compatibility result that makes the theory consistent with the literature. The existing VRP is not wrong; it is the special case at $a = 1$. It applies to formal-address markets, where it has been developed for seventy years. It does not apply to informal-address markets, where the general form is required.

10 Why You Cannot Buy This Off the Shelf

Commercial route optimizers implement the standard VRP. They can be used as subroutines for Stage 4 (sequencing) and Stage 5 (cross-cluster merge), but they cannot be used for Stage 1 (clustering) or Stage 2 (selection), because those stages require addressability data that the solvers do not have and would not know how to use if they did.

The build requirements are therefore:

1. **An addressability score** per place, maintained in the ledger and updated nightly. This is the input that commercial solvers lack.
2. **A margin function** per job that incorporates $\hat{p}$, $A(a)$, and $\mathbb{E}$[loss]. This is the objective that commercial solvers do not maximize.
3. **A clustering routine** that respects addressability variance. This is the constraint that commercial solvers do not model.
4. **A standard VRP solver** for the sequencing step, which is where the existing tools apply and should be used.

The build is modest. It is the first three items that matter, and none of them is available off the shelf.

11 Testable Predictions

1. Routes built by the contribution-maximizing algorithm outperform routes built by distance-minimizing algorithms, by a margin that grows as average addressability falls.

2. The optimal number of stops per route is higher in high-addressability corridors than in low-addressability corridors, because the variance buffer is smaller.

3. The optimal clustering is addressability-homogeneous: clusters with mixed addressability have lower achieved contribution than clusters with uniform addressability, at the same geographic dispersion.

4. The generalized savings algorithm produces more aggressive consolidation in corridors with high c_0 and high ΦC_p, and the effect is larger than the distance-only savings algorithm predicts.

5. The time-margin ratio ρ_j is a better predictor of job profitability than the margin m_j alone, because it accounts for the time cost of service.

6. In markets with $a \to 1$, the algorithm converges to the standard VRP solution, and the difference between them vanishes.

12 Conclusion

Route optimization in logistics is consolidation made operational. The objective is not distance, because distance is only one component of the time that a departure consumes. The correct objective is contribution per departure, and the correct constraint is shift length, and the correct input is addressability — because addressability determines both the time a stop consumes and the probability the stop succeeds.

The central result is simple: **the optimal route is the set of jobs with the highest margin per unit of service time, ordered by that ratio, included until the shift is full.** Everything else — clustering, sequencing, merging — follows from this rule, and the standard vehicle routing problem is the special case of this rule at perfect addressability.

This is why commercial solvers underperform in informal markets, and why the firms that used them without modification inherited their errors. The solver minimizes distance in a world where distance is not the binding constraint. The binding constraint is time, and time is dominated by addressability, and addressability is what the ledger records.

The route is the unit of competition. The margin-time ratio is the criterion. The ledger is the input. And the departure is the object that must be filled.

> **The theory in one sentence:** a route is a market for the driver's shift, jobs bid for it with their margin per unit of service time, the route fills with the highest bidders until the shift is full, and the standard VRP is what this looks like when every endpoint is perfectly addressed.

Algorithm 1 Contribution-Maximizing Route Optimization

Input: jobs J, cost parameters $(c_0, \Phi, C_p, c_{\text{km}}, c_r)$, shift length H, capacity Q, window W, confidence α
Output: set of routes $\{r\}$ and total contribution

Stage 1: Score each job
for $j \in J$ **do**
    $m_j \leftarrow \hat{p}_j A(a_j) n_j - c_{\text{var},j} - (1 - \hat{p}_j) c_r - \mathbb{E}[\text{loss}_j]$
    $\mu_j \leftarrow 1/R(a_j)$; $\rho_j \leftarrow m_j / \mu_j$
    if $m_j \leq 0$ **then**
        Remove j from J ▷ fails drop-level contribution
    end if
end for

Stage 2: Cluster by (corridor, addressability band)
Partition J into clusters $C_1, \ldots, C_k$ minimizing within-cluster addressability variance and geographic diameter

Stage 3: Select within each cluster
for each cluster C **do**
    Sort C by ρ_j descending
    $r \leftarrow \emptyset$; $T \leftarrow 0$; $Wt \leftarrow 0$
    for $j \in C$ in sorted order **do**
        if $T + \mu_j \leq H - z_{1-\alpha}\sqrt{\text{Var}[T] + \sigma_j^2}$ **and** $Wt + w_j \leq Q$ **then**
            $r \leftarrow r \cup \{j\}$; $T \leftarrow T + \mu_j$; $Wt \leftarrow Wt + w_j$
        end if
    end for
    if $G(r) > 0$ **then** emit r
end for

Stage 4: Sequence each route
for each emitted route r **do**
    Apply savings algorithm (Section 7) to obtain initial order
    Apply 2-opt / 3-opt local search to refine
    if $G(r) \leq 0$ **then** discard r ▷ sequencing revealed it unprofitable
end for

Stage 5: Cross-cluster merge (optional)
Apply generalized savings to pairs of routes from different clusters
Merge if $S(i, j) > 0$ and the merged route satisfies all constraints

return $\{r\}$ and $\sum_r G(r)$